

Math 104 – Real Analysis

Comprehensive exercise solutions

Ross, *Elementary Analysis: The Theory of Calculus* (2nd ed.) ■ Fall 2026 ■ Keshav Ramamurthy

Adding a solution

To record a solved exercise, type under its chapter banner:

```
\exercise{8.6} \begin{solution} ... \end{solution}
```

Restate the problem first with `\begin{problem}[title]... \end{problem}`; structure proofs with `claim`, `idea`, `proofsketch`.

Contents

Chapter 1	Introduction	2
Chapter 2	Sequences	3
Chapter 3	Continuity	4
Chapter 4	Sequences and Series of Functions	6
Chapter 5	Differentiation	7
Chapter 6	Integration	8
Chapter 7	Capstone	9

Chapter 1 Introduction

Ross, §§1–6: natural, rational, and real numbers; the completeness axiom

Example 1.1 (1.1). We proceed via induction. Let $a_n = \sum_{i=1}^n i^2$.

Then $a_1 = 1 = \frac{1}{6}(1)(2)(3)$, so the base case holds.

Then, we proceed via induction. Suppose $a_i = \frac{1}{6}i(i+1)(2i+1)$ for $i \in \{1, 2, 3, 4, \dots, k\}$.

We will show that this implies that $a_{k+1} = \frac{1}{6}(k+1)(k+2)(2k+3)$

We add $a_{k+1} = a_k + (k+1)^2 = \frac{1}{6}(k)(k+1)(2k+1) + (k+1)^2$ This is from our inductive hypothesis, and yields

$$a_{k+1} = (k+1) \left(\frac{1}{6}(k)(2k+1) + k+1 \right)$$

Pulling out $\frac{1}{6}$ yields $\frac{1}{6}(k+1)(k(2k+1) + 6k+6)$ And the second term of that polynomial is

$$2k^2 + 7k + 6 = 2k^2 + 4k + 3k + 6 = (2k+3)(k+2)$$

satisfying our inductive hypothesis.

Chapter 2 Sequences

Ross, §§7–16: limits of sequences, limit theorems, monotone/Cauchy, subsequences, limsup/liminf, series

Example 2.1 (2.2). Show that $\sqrt[3]{2}, \sqrt[7]{5}, \sqrt[4]{13}$ are not rational.

For the first for sake of contradiction set $\sqrt[3]{2} = \frac{p}{q}$ for $p, q \in \mathbb{Z}^+, \gcd(p, q) = 1$. Then, cubing both sides, we have $2 = \frac{p^3}{q^3} \rightarrow p^3 = 2q^3$.

However, since the rhs is divisible by 2, we have $2 \mid p \rightarrow p = 2p', p' \in \mathbb{Z}^+$.

Substituting this in yields

$$(2p')^3 = 2q^3 \rightarrow 4(p')^3 = q^3$$

So the analogous idea implies $2 \mid q$. However this is a contradiction, as $\gcd(p, q) = 1$ but we found 2 divides both p and q .

We can show the others analogously.

Example 2.2 (2.3). Show that $\sqrt{2 + \sqrt{2}}$ is not rational. Suppose for sake of contradiction we set $\sqrt{2 + \sqrt{2}} = \frac{p}{q}$ again, we assume $p, q \in \mathbb{Z}^+$ and $\gcd(p, q) = 1$. Squaring, we immediately have a contradiction because

$$2 + \sqrt{2} = \frac{p^2}{q^2} \rightarrow \sqrt{2} = \frac{p^2}{q^2} - 2$$

and the lhs is irrational but the rhs is rational.

Chapter 3 Continuity

Ross, §§17–22: continuous functions, properties, uniform continuity, limits of functions

Example 3.1 (3.1). For the set $\mathbb{N}$, we have: (for arbitrary $a, b, c \in \mathbb{N}$)

- a1: associativity holds $(a + b) + c = a + (b + c)$
- a2: commutativity holds. $a + b = b + a$
- a3: identity exists (0)

We continue and we see the existence of a multiplicative inverse, namely for $a \in \mathbb{N}$, $\exists a^{-1}$ s.t. $aa^{-1} = 1$ does not hold. For example consider $a = 2$ and $a^{-1} = \frac{1}{2} \notin \mathbb{N}$. $\mathbb{Z}$ similarly does not have a multiplicative inverse for all $a \in \mathbb{Z}$.

Example 3.2 (3.2). Commutativity of addition is shown in the proof (vi) to show $b^{-1} * ab = ab * b^{-1} = a$

It's shown in part (ii), while adding 0 at the beginning and then while adding to the rhs adding it to the right side (which is commutativity) then cancelling the zero out.

Shown in (iii), where $0 \cdot b = b \cdot 0 = 0$.

Example 3.3 (3.3). We wish to show that $ac = bc$ and $c \neq 0$ implies $a = b$. We have

$$ac + (-ac) = bc + (-ac) \rightarrow 0 = c(b - a)$$

Then we write

$$c^{-1} \cdot 0 = b - a \rightarrow 0 = b - a$$

Adding a on both sides yields $a = b$.

To show $ab = 0$ implies $a = 0$ or $b = 0$ for $a, b \in \mathbb{R}$

We proceed by contradiction. Suppose $a \neq 0, b \neq 0$. Notice we can write

$$a^{-1}(ab)b^{-1} = a^{-1} \cdot 0 \cdot b^{-1}$$

So the lhs is 1 but the rhs is 0, a contradiction.

Example 3.4 (3.4). To show $0 < 1$ we can suppose for sake of contradiction $0 \geq 1$. Then, we'd have $-1 \geq 0$, adding -1 to both sides Multiplying both sides by (-1) we'd have

$$(-1)(-1) \leq 0 \rightarrow 1 \leq 0.$$

Even though we used less than or equal to, obviously $0 \neq 1$ by definition, so we have a contradiction. Given the ordering properties, we wish to show if $0 < a$, then $0 < a^{-1}$.

Suppose for sake of contradiction $a^{-1} \leq 0$. Then we have, multiplying a on both sides

$$a \cdot a^{-1} \leq a \cdot 0 \rightarrow 1 \leq 0$$

which is a contradiction.

To prove $0 < a < b$ implies $0 < b^{-1} < a^{-1}$, we go in multiple steps.

We know that a^{-1} and b^{-1} are positive, we just wish to establish order. We multiply $0 < a < b$ by the constant a^{-1} to get $0 < aa^{-1} < ba^{-1}$.

Then multiply by b^{-1} to it to get

$$0 < b^{-1}aa^{-1} < ba^{-1}b^{-1} \rightarrow 0 < b^{-1} < a^{-1}$$

Chapter 4 Sequences and Series of Functions

Ross, §§23–27: power series, uniform convergence, term-by-term calculus

Chapter 5 Differentiation

Ross, §§28–31: derivative basics, Mean Value Theorem, L'Hospital, Taylor's Theorem

Chapter 6 Integration

Ross, §§32–36: Riemann integral, its properties, Fundamental Theorem of Calculus

Chapter 7 Capstone

Ross, §§37–38: exponents and logarithms; continuous nowhere-differentiable functions